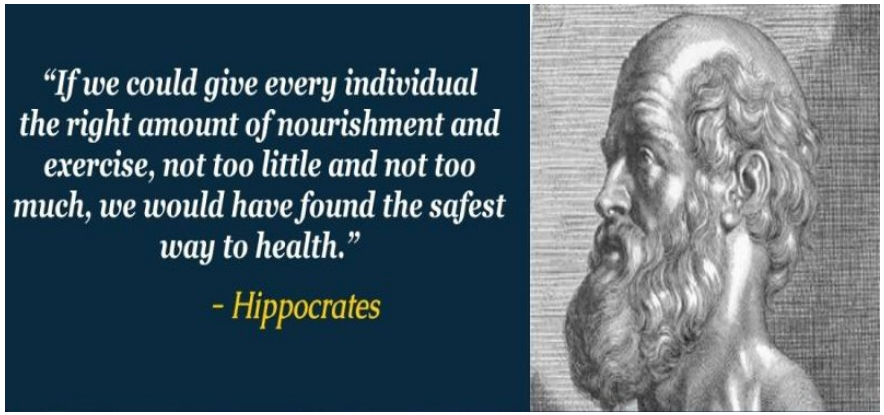
PHYT 6443 Hybrid Therapeutic Interventions
Module 2: Introduction to Therapeutic Intervention Prescription

Introduction to Physcal Rehabilitation



Did you know that physical rehabilitation techniques can be traced back to the time of Hippocrates?

Hippocrates was one of the first to document and advocate for things like aromatherapy, appropriate use of medication and surgical procedures, and various forms of movement to help the body heal naturally.

Physical therapist as Movement Specialists

The current APTA Vision Statement is: "Transforming society by optimizing movement to improve the human experience." While movement is only one piece of the puzzle, it is a big piece. Most of what you will do as a PT will be centered around helping people move more and body parts to move better. We do this through therapeutic exercises and manual techniques that we will cover in this class.

The purpose of this module and course is to help you learn the foundational principles for how and why various exercises/treatments work. This course will not teach you all there is to know about exercises, nor will it show you every exercise/treatment out there. However, empowered by foundational knowledge, you will be able to begin problem solving and creating your own treatment plans for your patents. The following pages will help you understand how to apply these prinicples and how far we have come in our understanding since the time of Hippocrates.

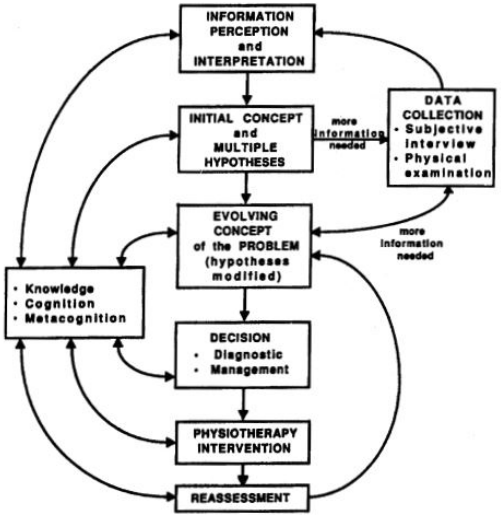
What is the value of prescribing therapeutic interventions?

"The ultimate goal of a therapeutic exercise program is the achievement of an optimal level of symptom-free movement during basic to complex physical activities." (Colby & Kisner, page 1, ed 7)

Patients and clients come into see a Physical Therapist (PT) because there is some part of their life that they are not able to fully participate in, but these activities are important to them. There is usually a physical impairment associated with these movement system disorders that is caused by injury, disease, or some health-related condition. Others seek out the advice of a PT because they would like to improve their overall fitness or quality of life while others seek advice for health promotion and wellness in order to reduce the risk of injury or disease. The PT is ideal for helping each of these groups of people because the PT is a movement specialist. PTs tailor therapeutic intervention programs to help the patient achieve this optimal level of symptom-free movement. These are best done by crafting a personalized treatment plan based on what the patient needs to improve their quality of life.

Using clinical reasoning to tailor therapeutic intervention programs

A little more than 1/3 of the time in PT school is devoted to foundational science, such as Anatomy, PT Exam Skills, EBP, Neuroscience, Integumentary, etc. As a clinician, you will use these foundation courses to know the different forms of exercise, how different exercises affect the different tissues of the body, how to appropriately load tissues for a desired goal, apply and integrate



This is a visualization of clinical reasoning processes that you will continue to develop as a PT. The more you work as a PT, the easier this process becomes. This graph will not be tested but it does help you understand the "process" of being a critical thinker in PT.

knowledge of anatomy, physiology, biomechanics, pathology, etc. across the broad spectrum of patient care. Additionally, your foundation classes help you know and understand the "why" and how things connect.

You will be able to use the foundations to help explain what may be causing the patient to have limited or symptomatic movements that are keeping them from achieving their goals, prioritize the impairments and functional limitations, and appropriately dose interventions to get the patient back to their goals. For example, a patient tells you they have issues reaching to get the spices out of the cabinet. You can use your foundation knowledge to determine that they have range of motion issues due to weakness in specific muscles. You would use your knowledge to help build a personalized program to strengthen the muscles that were weak.

Also, you have to look forward to the discharge goals, and come up with an effective, meaningful plan for how to get to these goals. PTs have to understand the relationships between physical functioning, health, and disability and how they relate to each patient's presentation. How a patient learns will impact how effective their plan of care is and needs to be considered when developing your treatments. You can develop the perfect plan of care but if the patient does not understand it or why it is relevant, then the outcome will be less desireable. Finally, you have to develop that patient-therapist alliance to create a team environment to get the patient back to what they love.

You will continue to develope these skills this semester. These principles will be further reinforced with every clinical class you take from here on out.

Defining Therapeutic Exercise

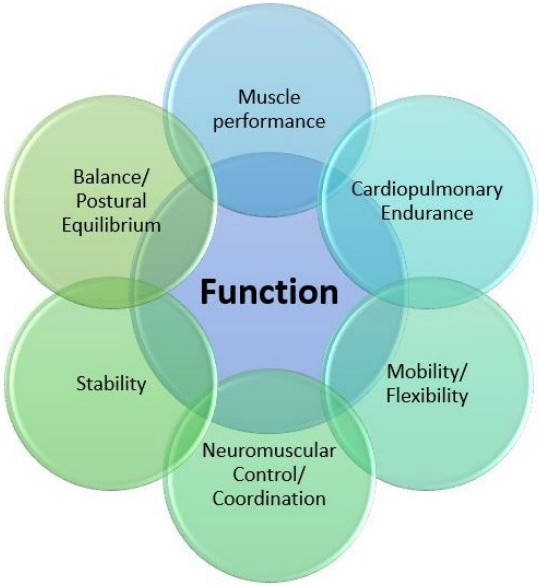
Therapeutic exercise is "the systematic, planned performance of physical movements, postures, or activities intended to provide a patient/client with the means to:

- Remediate or prevent impairments of body functions and structures.
- Improve, restore, or enhance activities and participation.
- Prevent or reduce health-related risk factors.
- Optimize overall health, fitness, or sense of well-being." Colby & Kisner, page 2, ed 7.

Big take-away points:

1. Systematic and planned means that we use clinical reasoning to determine the cause of the restriction and the best plan for how to fix it.
2. We do need to address impairments, but we always need to be sure we are assessing the functional participation. It is very easy to get caught up in the impairments, but that may or may not translate over into functional improvements. Think, "What does the patient have difficulty with in their daily lives?"
3. We are in a prime position working with patients to additionally address health-related risk factors. If we set up the patient-therapist alliance appropriately, we can make this happen well. An example is smoking decrease or smoking cessation.
4. Optimizing health, fitness, and sense of well-being are three umbrella terms with several related components beneath them. From a physiologic standpoint, we can certainly help with cellular and systemic functions as we help a patient return to function. Usually, functional participation involves appropriate movement; as we get the patients moving better without symptoms, their physical and emotional health is well-addressed. There are more than just health benefits that are addressed with well-reasoned therapeutic exercise prescriptions. Figure 1.1 below helps demonstrate the connection between systems and how function is at the center of the various systems.

Figure 1.1



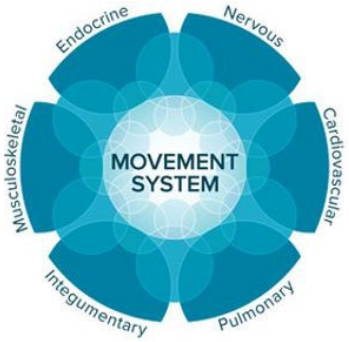
Remember, it is all about restoring the patient's ability to participate in functional activities that are important to them. This diagram breaks down some of the main components that are interrelated, which allows the patient to have that function. Disruption in any of these components will most likely affect other components and impair function.

Optional Brain Check

Task: Use the index cards below to refresh your understanding of the different components of Physical Function. These terms can be found on page 2 of your Colby & Kisner textbook.

The Human Movement Systems Using "Systems Thinking"

The human movement system is our foundation as a profession. It was a turning point in our profession to be labeled the movement specialist. This is why our goal in working with patients is to help them get back to the movements they need to be able to participate in the activities that are most important to them. The APTA's Vision Statement for our Profession is, "Transforming society by optimizing movement to improve the human experience," which reflects our specialty across all practices of Physical Therapy.



The human movement system is a balance of each of the systems of the body seen to the left. These body systems interact to control each of the elements of physical function that were on the prior page, such as balance, muscle performance, etc. Additionally, the body systems react, adapt, and develop in response to forces and physical stresses placed on the tissues that make up component parts of movement.

We will learn how to appropriately stress the body systems to improve physical function. Some stresses are good or normal. For example, gravity helps the body maintain functional levels of strength while gentle movements to a joint help it maintain mobility after a surgery. Using this idea of appropriate stress for the tissues in context of therapeutic exercise interventions allows us to have basis for selecting certain exercises at certain loads. We decide how much force and load to place on the human movement system, taking into account specific tissues or individual structures in a controlled environment. From there, we can progress or regress the exercises in a safe manner to improve movement, getting the patient better, faster.

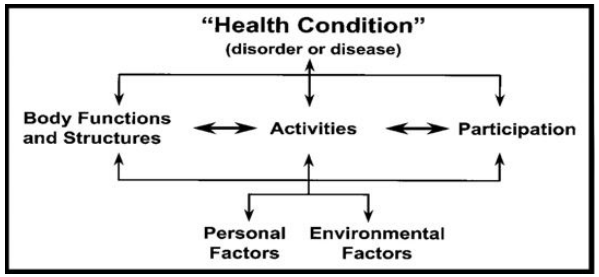
Inappropriate or insufficient stress to the Human Movement System

Inappropriate or insufficient stresses lead to impaired movement systems. Excessive load or forces can cause acute injuries, like sprains, strains, or fractures. Excessive loads can also cause chronic conditions, like stress reactions in bones or tendinopathies.

Insufficient loads can lead to degeneration, degradation, or deformity. An extreme example of this is astronauts in space. The lack of gravity and the long time that astronauts spend on the Space Station leads to significant muscle atrophy and loss of bone mineral density. NASA has had to come up with creative ways to help astronauts impose these stresses on their bodies in order to minimize damage.

Now that we have talked about being Movement Specialists addressing Movement System Impairments, let's revisit the ICF Model. If you need a refresher, please review your information from PT Exam Skills. This next page will put Therapeutic Interventions in context of the ICF Model.

Task: Click on the ICF model picture to the right, and it will link you to a YouTube video of a young college man who filled out his own ICF model. When you create your ICF model for your Therapeutic Intervention Project, it will be in much greater detail. This is just a reference point for where to start or see how one can use the ICF when getting a more full picture of the patient.



Knowledge Check: Catagorizing Elements of the ICF Model

Task: Review your PT Exam Skills information regarding the ICF model, and review Box 1.2 on page 6. The Brain Check below will ask you to match examples of the terminology used in the ICF Model. You can refer to pages 5-10 in the textbook for further details. Be sure you understand this part well because it will help with using the ICF to plan interventions.

Using the ICF Model to Guide Treatment

Converting the Diagnosis to Treatment

A key benefit of the ICF Model is that it allows the clinician to better understand the whole patient as an individual. Different patients have different goals, different activity limitations keeping them from their participation goals, and different impairments with different prioritization. Patients also have different environmental and personal factors affecting them, sometimes called the Social Determinants of Health. This helps us capture the PT diagnosis and prognosis.

With the emphasis on returning the patient back to their desired function, we can reflect and determine the different activity limitations that are causing the participation restrictions. Once the activity limitations are identified and prioritized, the impairments that are causing the activity limitations can be addressed. To address these, the PT may come up with 2-3 interventions to address the impairments. Once these are identified, the PT can plan the treatment effectively. This means that the treatment plan should be intentional and personalized which will potentially produce results quicker and more efficiently.

*The above box will be blank. It is a placeholder for the auto download.

It should auto-download, but if not, if you click on the link to the right. Dr. Vargo created a matrix to help you with deciding and prioritizing treatments when using the ICF model. You do not have to use this model, but you may find it helpful to build off of or to use to create your own. Keep in mind that you should not focus your whole treatment on addressing impairments. Once the impairment is safely addressed, you want to quickly work back towards the activity limitation and the participation restriction. Addressing the impairment does not necessarily mean that you are working the patient back to their goals. You eventually have to have activities that work back towards function.

Using the example from earlier, if a person has limited ROM due to weakness and their functional limitation is they cannot reach the spices in the cabinet. If you only focus your efforts on getting the arm stronger, then you would be missing the relation to function, aka. the reason why the patient came to you originally. Once they are would be strong enough, you would want to start incorporating reaching activities to help the brain connection and to strengthen functionally.

Notice how the gentleman can reach higher with good spinal posture. Poor posture leads to impingement in the subacromial structures earlier in the shoulder flexion ROM.

A few more points amount impairments to consider...

You may also see the terms primary impairment, secondary impairment, and composite impairment used. These are helpful terms to consider, especially with long-term therapeutic intervention planning.

Primary, or direct, impairments arise directly from a health condition or injury. Your textbook gives an example of a patient referred to PT with a medical diagnosis of rotator cuff tendinopathy. The patient might exhibit primary impairments of body function, such as pain, limited ROM, weakness of specific shoulder girdle muscles or glenohumeral muscles, etc.

Secondary, or indirect, impairments occur as a sequela to the primary impairments or occur in other body systems. The patient may present with postural impairments or faulty movement patterns of the head, upper extremity, or trunk that are caused by the pain and limited range of motion, not directly due to the tendon pathology.

Composite impairments are the result of multiple underlying causes (a combination of Primary and Secondary impairments). A patient may tear a ligament in their ankle and be immobilized for several weeks. They may demonstrate impaired balance due to chronic ligamentous laxity (body structure) and impaired ankle proprioception and weakness (body function) due to immobilization/disuse.

Why does this matter with my treatment planning?

Often times, treating the primary impairment can help the patient feel better in the moment. Only focusing on primary impairments does not focus on the root cause of the symptoms. The patient may plateau in their progress early on, and it can be frustrating for both the patient and the therapist!

Addressing secondary impairments early and concurrently helps address "the cause of the cause of impairments." By treating the underlying cause of the patient's main complaint, you are going to get them better faster, and you have a better chance of keeping the impairment from returning.

A good example is a patient that has low back pain secondary to hypermobility at L4-5 and a medical diagnosis of stenosis at that level. When watching the patient walk, you notice that they stand at their full height, but their pelvis remains in an anterior pelvic tilt. Additionally, the patient's hip does not extend well when they are walking. You do your evaluation, and you confirm hypermobility at L4-5. When assessing the hip, you notice decreased joint mobility in the hip, particularly with anterior glide (goes with hip extension, but you will learn this in Movement Science 1). If you try to stabilize L4-5 with all the right BP Cuff exercises, progression into weight-bearing and standing, address transitional movements, etc., you can get the patient's pain down. However, you are going to be more effective if you address the hip hypermobility at the same time you address the lumbar hypomobility. Without also treating the hip, the patient's symptoms will likely return. To improve efficiency, you can progress the treatment to address both of these regions with the patient standing and walking. This will improve the patient and help them return to their prior level of function by doing an exercise that is functional.

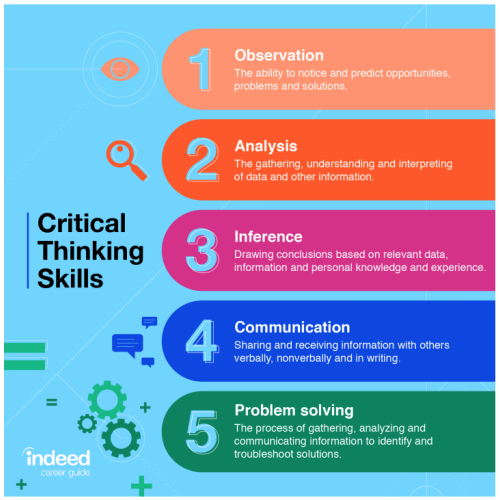
Key take-away: Think about how secondary impairments play into primary impairments, and try to treat both.

What is the difference between primary and secondary impairments and composite impairments?

A composite impairment happens when both primary and secondary impairments are the source of the composite impairments. The example from the textbook is the composite impairment of loss of balance. The case involves a patient who severely sprained the ankle in a traumatic injury, resulting in a torn anterior talofibular ligament. and subsequent follow-up with a walking boot. The primary impairments could be loss of proprioception, muscle weakness, and loss of ankle dorsiflexion from being in the boot. The secondary impairments would be increased lateral joint laxity, pain, and edema from the tear.

The Principles Behind Comprehensive Patient Management

To better understand the Patient Management Model, the principles that make up the model need to be reviewed because they are embedded in each phase of the patient management model. Be sure to review pages 12-14 (7th ed.) for further details and examples as needed.



Borrowed from: https://www.indeed.com/career-advice/career-development/critical-thinking-examples

Critical Thinking

Clinical decision-making is the art or critical thinking but in the clinical setting, typically with a patient focus. The infographic to the left puts critical thinking into steps or skills that help overcome challenges. These steps may be views as: 1) Observation, 2) Analysis, 3) Inference, 4) communication, and 5) problem-solving.

Using a real-life example, we can work through these. Example: you walk into a room and the lights are off.

1) what do you observe? - you notice the lights are of, you need light so this is a problem, the solution is to get light.

2) What data do you need to overcome the problem? - you could see if the light switch works, you could see if there is a bulb in the light fixture

3) What can you infer from the data gathered? If the light switch does not work, then you may draw the conclusion that there is an issue with the switch or wiring. If there is no light bulb, you know from personal experience that a bulb is needed to produce light

4) Who do I need to communicate about the issue to overcome the problem? you could reach out to your landlord, facility services, or the hardware store to determine in parts are available.

5) How do I overcome this problem? So far you have identified the issue, you have found the source of the problem, you have thought of a potential solution, you have communicated the problem, now you work to put it all together and to test out the problem through troubleshooting.

This is a bit of a basic example, but hopefully, this structured process can help you when you are stuck. If you apply this step-by-step thinking to patient care and competency exams, then you can process through any situation.

Clinical Decision-Making

The definition of clinical decision-making is the dynamic, complex process of reasoning and analytical thinking that involves making judgments and determinations in the context of patient care. In order to do this, therapists need to be able to merge clarification and understanding with critical and creative thinking.

An example that relates to this class is the misconception that PTs just need to follow a "recipe" of exercises. An example of this would be the common prescription of Bridges and Clams for any and all hip or knee issues. Using the process above, you can think about why a PT may have selected these exercises. Was it to target certain tissues? Was it to get the gluteal muscles to work without the back muscles doing all of the work? It is these clinical decisions that make our exercise prescriptions skilled PT services.

You may hear a PT talk about CPRs or CPGs. These are Clinical Prediction Rules or Guidelines. They are tools that we can use for specific issues like low back pain. The tool is a summation of evidence and gives the reader an idea of what treatment plans may be effective or less likely to succeed.

Box 1.7 Requirements for Skilled Clinical Decision Making During Patient Management

- Knowledge of pertinent information about the problem(s) based on the ability to collect relevant data by means of effective examination strategies
- Cognitive and psychomotor skills to obtain necessary knowledge of an unfamiliar problem
- Use of an efficient information-gathering and information-processing style
- Prior clinical experience with the same or similar problems
- Ability to recall relevant information
- Ability to integrate new and prior knowledge
- Ability to obtain, analyze, and apply high-quality evidence from the literature (EBP)
- Ability to critically organize, categorize, prioritize, and synthesize information
- Ability to recognize clinical patterns
- Ability to form working hypotheses about presenting problems and how they might be solved
- Understanding of the patient's values and goals
- Ability to determine options and make strategic plans
- Application of reflective thinking and self-monitoring strategies to make necessary adjustments

Optional Activity: The following is a sorting activity of examples of circumstances where it is appropriate for PTs to communicate and coordinate care of the patient with another provider.

Coordination, Communication, and Documentation

Coordination

As PTs, we are responsible for coordinating the care for patients who seek PT services. We decide if the patient would be best seen by a PT or PTA, whether a patient is appropriate for PT services, and when the patient is ready for discharge. In different settings, we work with a team to provide input on appropriate discharge locations, adaptive and assistive devices/technology needs, and even recommending other services or tests that the patient may need. We often will work on scheduling patients and managing the day-to-day flow of our schedules.

Communication

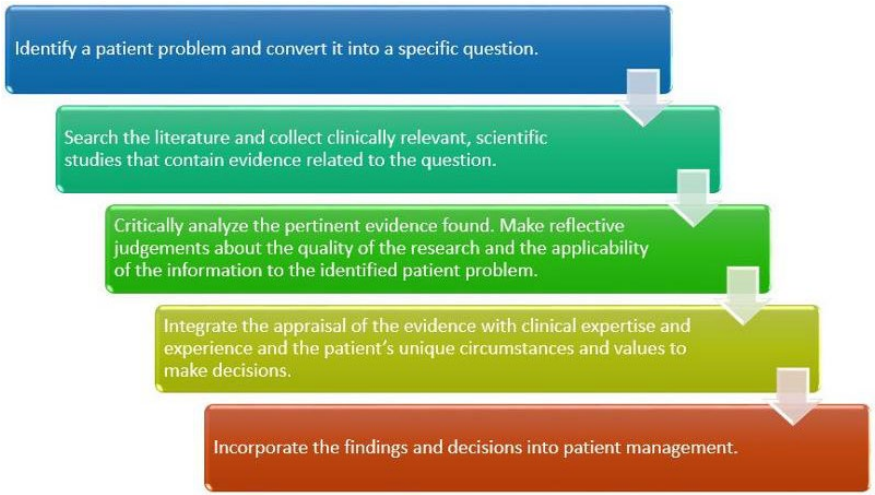
We'll review this more towards the end how this relates to patient education as well as you will cover more about teaching and learning related to patient care in the summer. The big thing to remember is to know who you are communicating with, what level of communication is appropriate (i.e. medical jargon may be appropriate for certain people), and how to follow up with written communication.

Documentation

Document thoroughly but concisely. Remember, if you don't document that it happened, then it didn't happen. Documentation should be efficient enough to get the point across but not overly verbose. Additionally, documentation should be timely. Document communications with other healthcare providers along with patient interactions. If you have a phone conversation, follow up with a written email summarizing the communication. The written follow-up allows for a reminder, a clear line of expectations, and a written document for people to fall back on if there are questions.

Evidence Based Practice

Below is a model using the principles of critical thinking and applying them to clinical decision making.



Key-Points: Notice how this involves delving into the literature and assessing the literature's applicability to your individual patient. You have to be a good consumer of the literature to know which articles best apply to your patient. EBP is not just about only using article-supported interventions. Rather, it involves balancing what is in the literature with your clinical experience and your thorough understanding of the patient sitting in front of you. There is no one size fits all in the PT world, but it makes each day in the clinic exciting and engaging!

The Patient Management Model

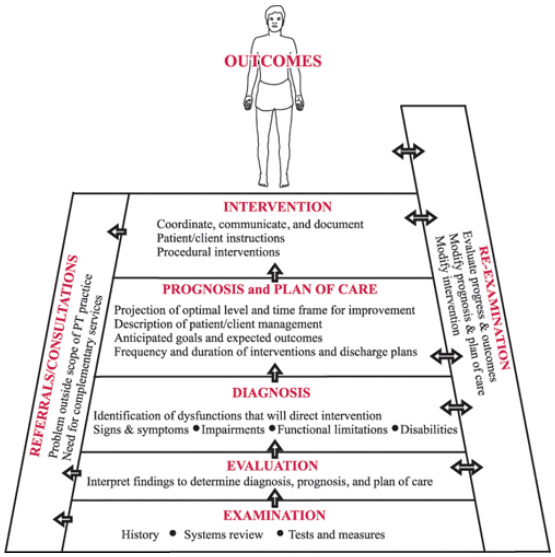


Figure 1.5. A comprehensive outcomes-oriented model of patient management.

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The model above helps demonstrate where in the treatment session certain components may fall. Recognizing that each fact gathered will help you progress futher into the model based on past experiences and clinical decision-making.

One more note about Diagnoses...

While the ICF Diagnostic Categories have not been adopted by the PT profession, it is highly recommended to use these Diagnostic Categories. You will need to tailor them to your patient, but below are some examples of these ICF Impairment Based Categories. We will add more to your "diagnostic toolbox" throughout this class and throughout your time here at UTMB.

- Examples of Primary Movement Impairments:
  - **Mobility Deficits**
  - **Movement Coordination Dysfunction/Deficits**
  - **Muscle Power Deficits/Force Production Deficit**
  - **Referred pain/Radiating pain**
  - **General debilitation/poor aerobic capacity**
  - **Sensory selection or sensory reweighting deficit**
  - **Fractionated movement deficit**
  - **Hypokinesia**
  - **Hypermetria**

The Patient Management Models: Focus on Interventions

An effective intervention may be described as one that results in reduction or elimination of body function or structure impairments, activity limitations, and/or participation restrictions. If you have used the ICF model to determine the most important impairments, limitations, restrictions, then the interventions can be best tailored to address the identified restrictions. Refer back to the tool created for you to see the lateral flow for using the ICF model.

While the use of any evidence-based model is a good idea, clinical decision-making must still be considered. In order to pick the best interventions, you need to best define the impairments and understand the goals associated with it. Look over to the right for determining goals related to impairments from the ICF model.

- **Mobility Deficits**
  - **Goal is to restore available range of motion**
- **Movement Coordination Dysfunction/Deficits**
  - **Goal is to restore optimal muscle performance,** motor synergy synchronicity
- **Muscle Power Deficits/Force production deficit**
  - **Goal is to restore muscle's ability to generate force, overcome/achieve functional load demands**
- **Referred pain/Radiating pain**
  - **Reduce sensory input coming from tissues**
- **General debilitation/poor aerobic capacity**
  - **Restore metabolic efficiency through oxygen delivery/substrate availability**

Once the goals are established that relate to the main dysfunction, what are next steps?

With goals identified, you will need to decide what interventions are best for the patient. The purpose of the intervention planning within treatment sessions and across treatment sessions is to bridge the gap between where the patient currently is functionally and where the patient/PT want the patient to be. It also involves constant assessments to determine if a patient is ready to progress, or if the desired outcome is not achieved, the patient needs to regress or change directions altogether.

Treating and resolving impairments alone does not mean that function is restored. Exercises need to progress to being task-specific or performed using movement patterns that closely match a patient's intended or desired functional abilities as the patient is ready. This means we need to consider a number of variables, including:

- Tissue quality
- Symptom severity
- Presence of co-morbidities
- Physical demands of ADLs and occupation

An example of task-specific training can be running. To run distance (anything greater than short burst), you ideally need strong legs. However, only working on strength, will not directly translate into optimal running performance.

We use our skills to be able to determine when progressions and regressions are appropriate

What type (Mode) of exercise do we prescribe?

This depends largely on the goal of the most relevant impairments (see above). Over to the right, there are examples of the types (modes) of interventions you might select based on the goals.

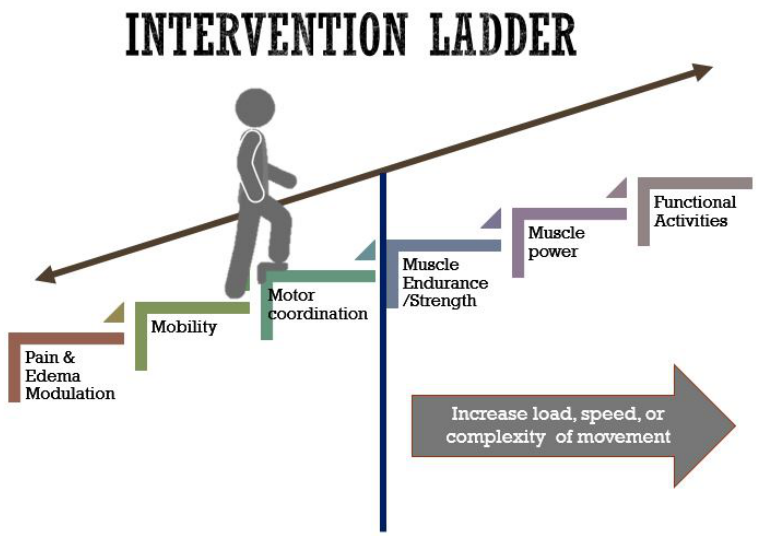
Remember, as the patient progresses or regresses, the impairments and goals will change. You need to know the modes of exercise based on the impairment goals, and how to progress and regress. That will be shown below.

For example: pt has mobility deficits -goal is to restore ROM. To increase range, you may do stretching. But if things hurt to much or are swollen, you may need to regress to a treatment that would address pain before stretching. Also, if passive stretching is to easy or the patient is not feeling it, it may make sense to do a stretch involving motor coordination aka an active stretch may be a good progression.

Hint: Notice how the color of the goal in this image coordinates with the color of the impairment highlighted 2 cells above it. Be sure to know how to match these back and forth.

- To increase joint **range of motion** (imbibition, **joint mobilization, stretching**)
- To increase **muscle flexibility** (**stretching, myofascial release/soft tissue mobilization**)
- To increase continuous **muscular output** (muscle **endurance**)
- To increase **muscle force generation** (**resistance/strengthening**)
- To increase/improve **motor recruitment** and skill acquisition (**motor coordination training**)
- To reduce **extracellular edema** (**tissue evacuation**)
- To reduce **pain** (**pain signal modulation**)

Key thing to remember: The Intervention Ladder for Progression



McGaugh, J. Exercise and Manual Techniques. UTMB, 2022.

Task: Save a copy of this Intervention ladder to refer to throughout this course and when you are in the clinic. This ladder shows the natural progression of types of exercises and interventions you should use when deciding the mode, intensity, duration, etc. of therapeutic interventions. Throughout each module where we discuss each of these steps, we will talk about indications and contraindications of each of these steps. We will also discuss how you know if the patient is ready to move onto the next phase/step. You will not learn every exercise, manual therapy technique, or functional activity in this class. You will learn how to progress or regress these interventions along lines of this ladder throughout the course, which will highlight the clinical decision-making skills you are developing!

Putting Concepts into Practice

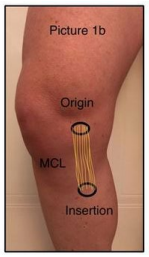
Case Study

This is for your practice to help you along your understanding as you progress in the course. Test material will not come from this page

A 27-year-old graduate student was referred to Physical Therapy with acute right knee pain and swelling after an injury to the knee 1 week ago. The patient shares that she was performing a small jump dance move and landed in an awkward way that caused her fall. As part of the fall, she experienced an acute subluxation/reduction of her patella and a grade II injury to her MCL. Her knee is now painful, swollen, and she is not able to walk without crutches.

The patient is an active female, and she exercises about 3-4x/week. She enjoys jogging and weightlifting. She has no prior medical history to note, no significant family medical history, and there is no chance that she is pregnant at this time. She denies smoking or illicit drug use. She has about 2 glasses of white wine per week. She has access to her program's fitness center, which is open from 6:00 am - 10:00 pm daily.

The student reports that she is extremely concerned about her success in her graduate program. She reports that this knee injury has caused her to lose a significant amount of time that she should be using to study for her qualifying exams because of all of the appointments she has had to attend. She also has difficulty sitting for prolonged periods of time needed for studying.



Examination

Evaluation

Due to the acuity of the injury, you find that the patient is now presenting with the following findings:

1. Edema, which will produce loss of muscle power, motor coordination, and ROM
2. Pain, result of edema and tissue injury/intensity from the ligament tear
3. Decreased ROM, due to edema
4. Movement coordination dysfunction, due to edema, pain, altered muscle power, and altered mechanoreceptor input. If we look at secondary impairments, there is a strong correlation of patellofemoral subluxations and knee valgus injuries that occur in females, especially with hip motor coordination dysfunctions.
5. Decreased muscle power, due to edema, pain, and loss of ROM
6. Decreased balance and proprioception, due to edema, tissue injury, and altered mechanoreceptor input
7. Activity limitations with gait, transfers, bed mobility, showering/bathing, lifting
8. Participation restrictions in accessing living quarters, accessing classrooms and university services, driving, returning to dance, possibly intramural participation, possibly other recreational physical activity

Diagnosis

There are a lot of potential diagnoses that you could write. Here is an example:

S/P traumatic grade II MCL sprain and patellofemoral subluxation producing mobility deficits.

Why mobility deficits? For right now, based on the acuity of the injury, the edema is causing pain, decreased ROM, motor coordination dysfunction, muscle power deficits, and impaired balance. A lot of this can clear up quickly once the acuity of the injury passes and once we get that swelling out of the knee to restore motion. It would also allow the patient to have decreased pain and tolerate more functional weight-bearing.

Could you have picked motor coordination as the primary deficit? Yes, it would depend on your reasoning. Knowing that the cause of the cause comes from what happened at her hip and knee with jumping, you could go down the motor coordination route. You could justify that motor coordination may improve once edema decreases, and focusing interventions quickly after that on motor coordination at the hip and knee will get the patient more quickly back to function.

Could you have picked muscle power deficits? You could, but the reasoning is not as strong for right now given the acuity of the injury. To be clear, this patient's primary impairment will quickly shift to muscle power deficits once you clear up the edema and motor coordination so that you can get her back to her optimal level of function.

What I hope you take away from this point is that you can have more than one right answer. The more you develop your clinical reasoning, the easier it is to justify your reasons. The "why" aka why you choose to do something matters and should be directly related to the patients impairments.

As a general rule of thumb, taking care of the acute nature of something will take priority over working on potential causes. Example, if the whole room is on fire, putting out the candle that started it doesn't not matter at that moment but once you get the room under control, you can turn your attent better to the source of the issue. Sometimes you can go to the source to stop it from making the situation worse or making it continue, but often the acute nature of pain and edema will prevent you from effectively treating the cause. Thus you may find that treating the acute symptoms first will allow for a more successful treatment of the cause. Once the initial phase of acuity has gone down, aka the swelling has reduced and pain is manageable, you may find yourself aiming to treat the cause of the issue first and any treatments for symptoms may be used more as an adjunct to your primary treatments.

Keep in mind that we are holistic providers and should consider other resources the patient may benefit from. In this case, it may also be a good idea to refer the patient to a place like the Office of Academic and Student Affairs for accommodations in classroom settings, accessibility, etc. while the patient is recovering and still has gait deficits.

Prognosis and Plan of Care

Given the acuity of the injury, there has not been time for chronic tissue changes to set in. The tissues are acting like normally inflamed tissues. The patient has an active lifestyle with no significant medical history to report. She has access to fitness facilities, and she has a higher education level. Her prognosis is good at this point.

What is her current functional status? She is ambulating with crutches with an edematous knee secondary to the injury.

What is her goal functional status? She wants to be able to fully participate in classes and return to her prior level of physical activity.

How do we get her there? Remember, this is going to depend on the impairment classification you selected. Based on her participation restrictions and her activity limitations, I had decided at this point in time that the mobility restriction was her primary impairment. Therefore, I would plan to initially focus on improving joint and soft tissue mobility to decrease edema, decrease pain, and improve the muscle pumping action with ROM exercises. This is also the bottom of the intervention ladder, which means it needs to be met first. Once this improves and clears out, my primary impairment would quickly change to motor coordination deficits, and I would plan to focus my interventions on motor coordination.

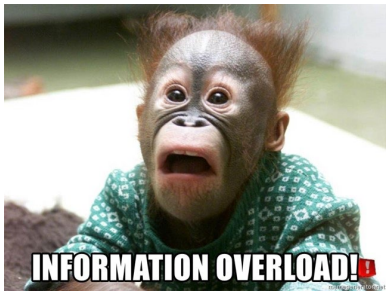
What would be some goals that you would write? You may post examples in the Discussion Board, and provide feedback on your classmates' goals.

Intervention

We will begin to focus on different intervention strategies. You may begin to think of what you would do based on the issues you wanted to focus on. Challenge yourself to create a progression and regression for any exercise you choose. Try to increase/reduce difficulty in the same area of impairment. In other words, if the impairment is balance, adding weight-resisted exercises would not be directly related to balance. Be sure to review the Impairment Classifications from the ICF model, the goals for impairment classifications, the general treatments for the goals, and the Intervention Ladder.

Suggested Break Time

There is a good amount of information to process through. Be sure to take breaks to help make sure you understand the material and not simply know it at a surface level.



Converting your Knowledge into Meaningful Movement

Even Dr. Shirley Sahrmann, the founder of the Movement System Impairment Syndomes, (you will cover these in ortho classes) often has questions about how long the patient should continue their exercise program. In short, the movements and interventions we prescribe are not (typically) just a short-term solution but will require life/behavior change.

So how do we help our patients unlearn faulty movement patterns or habits, learn new ways to move, and keep moving in this way? This is a multifaceted question that we will touch on but will continue to be addressed in many future classes but first we help the patient by helping them learn about their health.

This is the pdf of the results of the National Assessment of Adult Literacy. This is **"nice to know"** information, but it is **NOT "need to know"** information. You will not be tested over the contents of this pdf. It is here more for you to have access in case you want to look further into health literacy.

If this does not auto-download, you can find the file here:

https://nces.ed.gov/pubsearch/pubsinfo.asp?pubid=2006483

Health Literacy

Health literacy, as defined by the National Library of Medicine, is "the degree to which individuals have the capacity to obtain, process, and understand basic health information and services needed to make appropriate health decisions." This is deeper than literacy because it requires the ability to make decisions based on information presented.

The most recent assessment of Health Literacy in adults in the United States was in 2003. Unfortunately, 89% of subjects fell below "Proficient." Fortunately, 53% of subjects fell into the "Intermediate" category, which includes factors like ability to determine a healthy weight range according to BMI, ability to determine which vaccines are needed at a given age for children based on a chart, and identifying contraindications for medications based on an information pamphlet. These subjects are able to be safe, assuming information is given to them, but is the level of health literacy enough to truly make decisions? Between 2010 and 2014, additional data was collected to determine general literacy for the United States. In their findings, Texas is below average for both literacy and numeracy compared to the nation.

As PTs, we cannot overhaul health literacy on our own throughout the entire country. However, we can and should consider health literacy factors as we develop our teaching and communication skills to tailor our instructions to each patient. We need to understand each patient's health literacy level when we provide verbal and visual instructions to patients. We need to communicate in plain language, provide clear and user-friendly handouts/brochures/videos/pictures, and be mindful of our feedback to patients during sessions and between sessions. We need to constantly strive to close that gap between what we know as professionals and what the patient understands. This gives the patient ownership in their health, and it leads to longer-lasting benefits.

A common area we see issues with health literacy is related to pharmacological management of issues. Polypharmacy is unfortunately common and at the core is the assumption that patients know why they are taking medications, how to take them, what side effects to report, and what interactions may take place. While not our primary career, you may be the one who helps a person understand the importance of a medical thing (drug, surgery, procedure, etc). Don't assume someone understands, always ask and adjust as needed.

Building Rapport and Success

Remember that an effective plan of care strongly benefits from a patient-therapist alliance (commonly called the Therapeutic Alliance). You will help the patient better achieve their goals by:

1. Directing care towards goals and explaining where the treatment fits in the plan to get the patient back to their goals;
2. Understanding the patient's learning style and targeting that style with instruction;
3. Understanding the patient's perceptions and attitudes towards exercise to determine how receptive the patient is to learning about and adhering to exercise (The behavior-change model);
4. Start and end with success in mind. Begin with easier exercises that the patient can achieve, and end a treatment session with success. This will help with motivation carry-over. Keep in mind a person's perception. If they don't understand why, if they have excessive pain, if they can't succeed, they will become frustrated and will not do as well no matter how "perfect" the plan of care.

For all the pedi people in the room, your patients can be a tough crowd! Kids read people very well, and they can be brutally honest!

Task: Read Box 1.18 in your textbook on page 28 for practical ways to incorporate successful exercise instructions.	

Converting your Knowledge into Meaningful Movement: Introduction to Motor Learning

Introduction to Motor Learning

Motor Learning is different than Motor Performance. While Motor Performance means that the person has or develops the ability to carry out a skill, Motor Learning involves both the ability/development of ability to perform the skill as well as retain the skill. Motor learning involves complex internal processes that allow a person to perform and retain the ability to perform a skill or movement throughout practice. There are true neurologic changes that happen in the central nervous system that organize, process, and affect how motor actions are produced. In a very simple explanation, it is how we develop habits of movement or skills.

You will learn a lot more about this in Movement Science 2. The purpose of introducing these concepts in this course is to help you incorporate the basic principles into your intervention prescription and patient education. This will help you develop your patient care skills.

Types of Motor Tasks

There are three different kinds of tasks that can be performed. Practice identifying the types of tasks to the right.

Discrete Tasks: These tasks involve an action or movement with a recognizable beginning and end.

Serial Tasks: These tasks involve a set or series of discrete movements that are combined in a particular sequence.

Continuous Tasks: These tasks involve repetitive, uninterrupted movement with no reasonably distinct beginning or end. Although the action may be stopped because of fatigue, the task in and of itself involves repetitions of the same movement.

Being able to recognize the type of task a patient needs to perform can help a therapist decide the best teaching methods for that specific task, pending the stages of motor learning and variables in motor learning described below.

Conditions and Progressions of Motor Tasks

We do not necessarily perform the same task the same way every single time. Therefore, if we truly learn a skill, we have to be able to perform it in a variety of conditions. We need to anticipate these conditions and safely progress the patient with increased complexity of the tasks. Below is a representation of the taxonomy of motor tasks. Notice how the top left corner involves the most simple task with increased stability, no variability, no manipulation, and in a controlled environment. Changing the complexity of any of the four task dimensions makes a task more complex. You want to be sure you progress the patient, but at a rate that is appropriate for the patient.

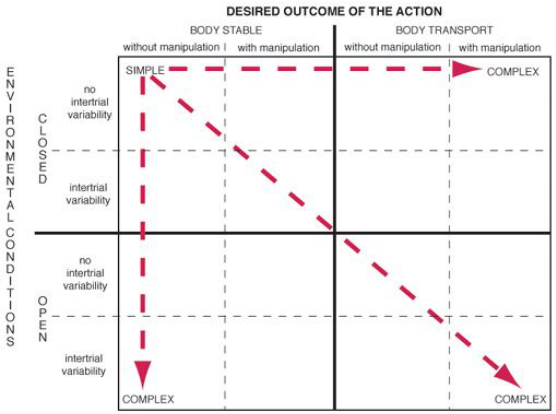


Figure 1.8. Taxonomy of motor tasks: dimensions of task difficulty.

(From Dennis, JK, McKeough, DM: Mobility. In May, BJ [ed]: Home Health and Rehabilitation—Concepts of Care. Philadelphia, FA Davis, 1993, p 147, with permission.)

Hint: Be sure you understand how to change the task dimensions to make a task more or less complex!

4 Main Task Dimensions

Review the Cliff's Notes version of the Task Dimensions below. Refer to pages 28-30 in your textbook (7th ed) for more details/clarification. Then, complete the Quiz Popper to classify the task conditions.

Environment

- Closed: Objects around a patient and the support surface do not move. This is a more simple task.
- Open: Objects or other people are in motion AND/OR the support surface is not stable. This is a more complex task.

Intertrial Variability

- Absent: The environment/condition in which a task occurs is constant (unchanging) from one performance to the next. This is a more simple task.
- Present: Demands in the environment/condition change from one performance to the next. This is a more complex task.

Body Stability

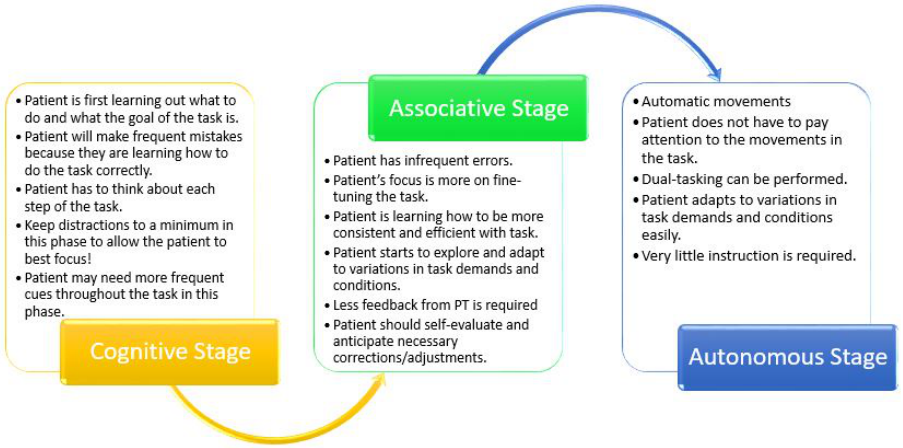
- Body Stable: The patient keeps their body in a stable, stationary position. This is a more simple task.
- Body Transport: The patient is asked to move from one place to another. This is a more complex task.

Object Manipulation

- Absent: The patient is not asked to manipulate an object, which leaves the hands, and possibly feet, pending the task, free to focus on the task requested. This is a more simple task.
- Present: The patient is asked to hold, carry, kick, manipulate an object. This is a more complex task.

Have a good understanding of these. They will show up throughout the curriculum because a PT uses these concepts to make things more complex or less. There can be a multiplying effect if you change to many things at once. Typically, you would want to only change one at a time and then assess how the patient did. Sometimes a change in one area (variability for example) may feel like a big change for the patient while the same level of change in a different category (environment for example) might only be perceived as minor. This is why it is important to make only one type of change and then assess.

Stages of Motor Learning



These are some brief details that you will want to consider with the Stages of Motor Learning. It will help you consider how much feedback is necessary and when a patient is ready for the next level of complexity. Click on the link below to a YouTube video that summarizes the same concepts. It gives some visual cueing that you may find helpful.

Motor Learning from a PE Coaching Perspective

Perfect Practice Makes Perfect

Task: Read the section in your textbook regarding "Practice" on pages 31-32. In particular, make sure to know how these types of practice can be used with therapeutic interventions and the appropriate place for these types of practice.

Key Points to Remember

- **Part practice** is appropriate in the **early learning stages** of **complex serial skills** with simple and difficult components. It is also appropriate if **the patient's cognitive status** requires more skill breakdown.
- **Whole practice** is more effective for learning **continuous tasks, serial tasks that require momentum or specific timing, and discrete tasks**.
- At the **early stages** of motor learning, **blocked-order practice** is most appropriate because it can lead to rapid improvements. It needs to quickly transition to random- or random/blocked practice.
- **Blocked practice** may still be the method of preference if a **patient's cognitive ability** is too challenged by random practice, which would interfere with learning.
- **Random-order practice** leads to **better skill retention and generalizability** than blocked practice.
- **Random/blocked order practice** results in **faster learning** than random-order practice and **better retention** than blocked practice.
- **Mental practice** used **in conjunction with physical practice enhances motor skill acquisition** at a faster rate than use of physical practice alone.

Feedback

The key points below are general statements. You should also encourage a patient to provide input about their receptiveness to the type of feedback or feedback schedule used during practice, especially after they get in their groove of skill acquisition. Some patients request a lot of feedback, and some despise feedback. You can work with them to get them to the recommended schedule/type of feedback, but it may be more of a process with some patients than others.

Task: Read pages 33 and 34 in your textbook for further definitions/examples of feedback. Understand the benefits and limitations of the types and schedules of feedback for acquisition and retention of skills. Read Box 1.22 on page 36 for examples of how to incorporate appropriate feedback at the different phases of motor learning.

Key Points to Remember

- **Intrinsic feedback** only involves the patient's sensory systems. There is no feedback from the PT. The **benefit** is it can provide feedback in the moment or at the end of a task, and it **is inherent to everyday life**. The **limitation** is the **patient may receive inadequate or impaired feedback** and require augmented feedback.
- The **therapist provides augmented feedback**, which can have several tactile, verbal, visual, or auditory forms. The PT controls the type and schedule of the feedback. Another benefit is that it **enhances intrinsic feedback, especially with impaired internal feedback**. The **limitation** is it **requires another source of feedback**, and **it needs to be tapered down**. Remember, "less is better."
- **Knowledge of Performance** allows patients to recognize and correct components of skills. The **limitation** is that it **does not promote retention of a skill as well** as Knowledge of Results.
- **Knowledge of Results produces better retention of a skill**. The limitation is that a faulty movement or habit may continue to be show up if not corrected.
- **Concurrent feedback** results in **quicker skill achievement**, but **less effective skill retention** that immediate post-response and summary feedback.
- **Summary feedback** produces **better skill retention** than concurrent feedback and immediate post-response feedback, but the patient has **slower skill acquisition**.
- **Variable, or intermittent, feedback results in better retention** than constant (continuous) feedback, although **constant feedback** results in **better skill acquisition** at the beginning stages of motor learning.

- Please note that this is a spectrum. In no way should you only provide knowledge of results as a summary feedback for every patient. It depends on where the patient is in the motor learning process and how the feedback best supplements their learning styles. You will help them progress to the ideal forms of feedback appropriate for where they are in the learning process.

Short Brain Check and a Suggested Break

Guidelines for Creating an Intervention Prescription

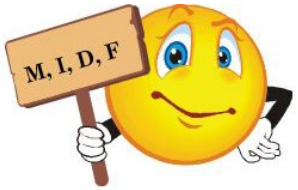
Once you have gone through your clinical decision-making to determine the participation restrictions, activity limitations, main impairments, then created a diagnosis and prognosis, and finally, determined the tissue involved and its state of healing, you are now ready to pick your specific interventions. As you need to at this stage, refer back to the ICF flowsheet for planning your interventions. You will want to consider Outcome Goals and how the interventions relate back to function.

The overall theory related to this intervention prescription is the Physical Stress Theory. We want to appropriately stress the tissues to promote the best healing environment for the tissues. Determining intervention dose requires making choices about variables that apply stress to the tissue. For each intervention, you will want to decide the M, I, D, F.

Mode: (what type of intervention)
Intensity: (amount of force necessary to achieve the activity)
Frequency: (how often the exercise is performed)
Duration: (number of repetitions or time the exercise is performed)

You have a couple of more decisions to make regarding your intervention. These decisions will be based on the mode of exercise you select, as well as observations you make of the patient demonstrating the exercise.

- **ROM**: how much tensile stress? how much muscle effort?
- **Type of muscle contraction**: isometric, concentric, eccentric
- **Speed of movement**: adds a motor coordination element
- **Quality of movement:** is there a muscle imbalance problem?



Example of MIDF:

M: Squats

I: Body weight

F: 2x a day, 4 x a week

D: 5 set x 8 reps each

Using the above example, how can you play with the ROM, contraction type, speed, and quality?

Revisiting the Physical Stress Theory

Exercise volume: The total amount of exercise performed in a single session.

We need to look at more than just the exercise volume, especially in context of the ICF Model. We need to consider **exercise volume + extrinsic factors + psychosocial factors + physiological factors** when dosing out our interventions.

What is our goal with the tapping into the Physical Stress Theory?

- To provide safe and sufficient volume of physical stress to produce tissue adaptation without causing tissue injury or death.

Brody LT. Effective therapeutic exercise prescription: the right exercise at the right dose. JHT 2012.

Tying it Back Together

The Physical Stress Theory produces Adaptation Responses.

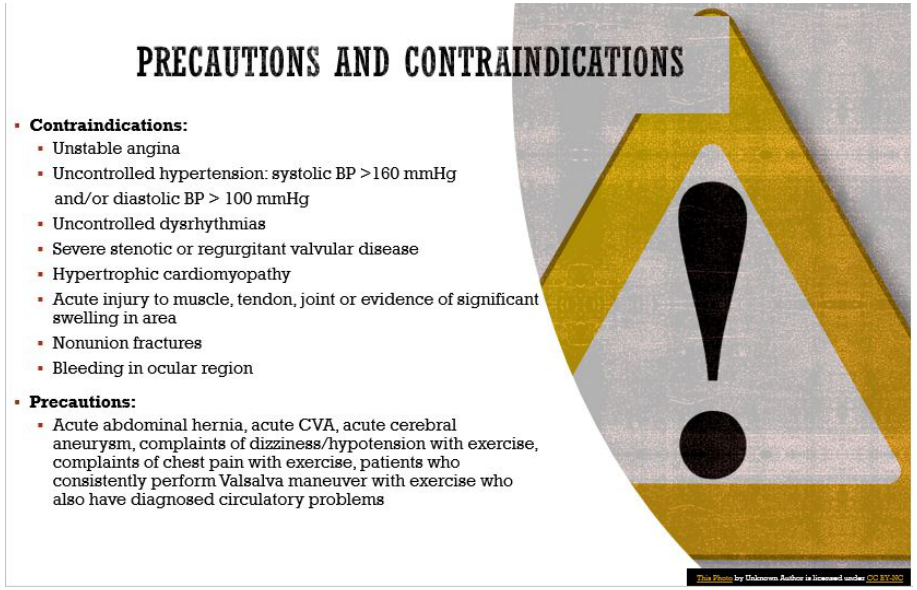
Exercise-induced adaptation is:

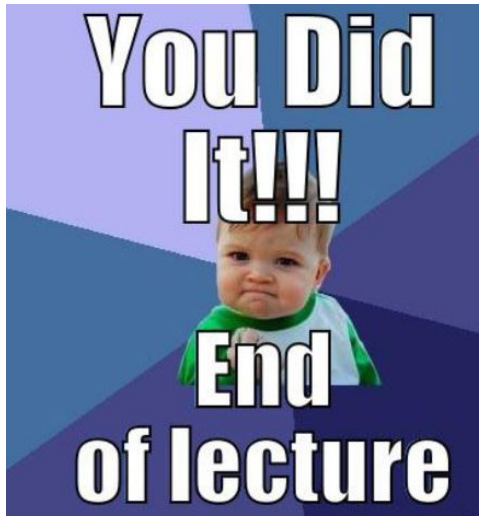
- Complex process mediated through **changes in muscle cell signaling**
- Determined by **training volume, intensity, and frequency** in addition to protein half-life
- **Specific to mode** of training

To bring this full circle and where to move forward, we have discussed in this lesson why specific modes of training are selected with the SAID Principle, tissue healing, patient success, and return to specific function. We have talked about training volume, intensity, and frequency in broad terms, as well as ensuring that signs of inflammation do not persist or worsen. We will get into more specifics as we progress through the primary impairments and related modes of interventions. On the next page, we are going to talk about the changes in muscle cell signaling.

General Precautions and Contraindications when Considering Exercise Prescriptions

You need to know these precautions and contraindications in terms of identifying them and recognizing how they might play into a case scenario. I do NOT expect you to understand this at the level that you will this summer in your Cardiopulmonary class. For now in this class, recognize them as contraindications (warrants further follow-up with MD, withhold exercising until clear communication with MD) or precautions (keep the mode and intensity light, monitor vital signs closely). While you do not need to know the why behind something being a precaution/contraindication at this time, you do need to make sure you know what each thing means. For example, if you have forgotten what hypertrophic cardiomyopathy means, you may wish to search what that means but you do not need to know the ins and outs of causes.




You Did
It!!!
End
of lecture